WORKED SOLUTION

STEP II Specimen paper, Question 12

Mechanics • Oscillations on a Wire

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Read the question before you go on. It is in the STEP Support Programme's question database:

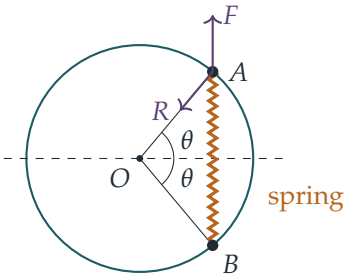
step.maths.org/questions/spec-s2-q12

The key idea

The wire is smooth, so the only forces doing work are the spring's, and mechanical energy is conserved. All we need is the length of the spring in terms of θ (a chord of the circle) and the speed of each bead. Two things need a moment's thought. Why does the line bisecting angle AOB stay fixed? And in part (ii), what happens if the beads have enough energy to reach opposite ends of a diameter?

Solution

Model the beads as particles on a smooth circular wire in a horizontal plane, joined by a light spring obeying Hooke's law. Gravity is vertical and is balanced by the vertical part of the wire's reaction, so it plays no part in the horizontal motion.



In the diagram, F is the spring's push on A (along BA, since the spring starts compressed) and R is the horizontal part of the wire's reaction on A.

Setting up the energy equation

Measure each bead's position by its angle on the circle: A at angle ϕ_A and B at angle ϕ_B.

Ask yourself Why can we assume both beads move with the same speed, symmetrically about a fixed line?

Take moments about the vertical axis through O. The spring forces on A and B are equal and opposite and act along the same line AB, so their moments cancel. The horizontal part R of each wire reaction is perpendicular to the wire, so it points along the radius through O and has no moment. The vertical forces have no moment about a vertical axis. So the angular momentum $ma^2(\dot\phi_A + \dot\phi_B)$ is conserved. It starts at zero, since the beads are released from rest, so $\phi_A + \phi_B$ stays constant: the bisector of angle AOB is fixed. Take it as the dashed line in the diagram, so that $\phi_A = \theta$ and $\phi_B = -\theta$. Then each bead moves with speed $a\dot\theta$.

The spring is the chord AB, of length $2a\sin\theta$. Its natural length is $2a\sin\alpha$, so its extension is $x = 2a(\sin\theta - \sin\alpha)$, and its elastic energy is

$$\frac{\lambda x^2}{2\cdot 2a\sin\alpha} = \frac{4\lambda a^2(\sin\theta-\sin\alpha)^2}{4a\sin\alpha} = \frac{\lambda a(\sin\theta-\sin\alpha)^2}{\sin\alpha}.$$

This formula also holds when $x < 0$ (compression), because x is squared. The kinetic energy is $2\cdot\frac{1}{2}m(a\dot\theta)^2 = ma^2\dot\theta^2$. The wire is smooth and its reaction is always perpendicular to the motion, so it does no work, and energy is conserved. At release $\theta = \beta$ and $\dot\theta = 0$, so

$$ma^2\dot\theta^2 + \frac{\lambda a(\sin\theta-\sin\alpha)^2}{\sin\alpha} = \frac{\lambda a(\sin\beta-\sin\alpha)^2}{\sin\alpha}.$$

Multiplying through by $\sin\alpha/a$ gives

$$ma\dot\theta^2\sin\alpha + \lambda(\sin\theta-\sin\alpha)^2 = \lambda(\sin\beta-\sin\alpha)^2,$$

as required. Note that the spring starts compressed, so $\beta < \alpha$.

Part (i)

Differentiate the energy equation with respect to t:

$$2ma\sin\alpha\,\dot\theta\ddot\theta + 2\lambda(\sin\theta-\sin\alpha)\cos\theta\,\dot\theta = 0.$$

Dividing by $2\dot\theta$ (valid whenever the beads are moving, and the result extends to all t by continuity) gives the equation of motion

$$ma\sin\alpha\,\ddot\theta = -\lambda(\sin\theta-\sin\alpha)\cos\theta.$$

The equilibrium is at $\theta = \alpha$, where the spring has its natural length. If $\beta - \alpha$ is small then θ stays close to α (the energy equation gives $|\sin\theta - \sin\alpha| \le |\sin\beta - \sin\alpha|$), so write $\theta = \alpha + \varepsilon$ with ε small. To first order, $\sin\theta - \sin\alpha \approx \varepsilon\cos\alpha$, and $\cos\theta \approx \cos\alpha$, which multiplies a quantity that is already small. So

$$\ddot\varepsilon \approx -\frac{\lambda\cos^2\alpha}{ma\sin\alpha}\,\varepsilon,$$

which is simple harmonic motion with $\omega^2 = \lambda\cos^2\alpha/(ma\sin\alpha)$. The period is therefore

$$T = \frac{2\pi}{\omega} \approx 2\pi\sqrt{\frac{ma\sin\alpha}{\lambda\cos^2\alpha}}.$$

Part (ii)

Write $k = \sqrt{ma\sin\alpha/\lambda}$. From the energy equation,

$$\dot\theta^2 = \frac{1}{k^2}\Big[(\sin\alpha - \sin\beta)^2 - (\sin\theta - \sin\alpha)^2\Big].$$

The beads stop when $\sin\theta - \sin\alpha = \pm(\sin\alpha - \sin\beta)$, that is when $\sin\theta = \sin\beta$ or when $\sin\theta = 2\sin\alpha - \sin\beta$. Starting from $\theta = \beta$, θ increases (the compressed spring pushes the beads apart), and $\dot\theta^2 > 0$ for as long as $\sin\beta < \sin\theta < 2\sin\alpha - \sin\beta$.

? **Ask yourself** What decides which of the two cases we are in?

Whether θ can reach a second stopping point before $2\theta = \pi$, where A and B are at opposite ends of a diameter and $\sin\theta$ reaches its largest value, 1. That depends on whether $2\sin\alpha - \sin\beta$ is less than or greater than 1.

(a) $\sin\beta > 2\sin\alpha - 1$**.** Then $2\sin\alpha - \sin\beta < 1$, so there is an acute angle γ with $\sin\gamma = 2\sin\alpha - \sin\beta$, and $\gamma > \alpha$. The beads come to rest when $\theta = \gamma$ (the spring is then stretched by as much as it was originally compressed), and θ oscillates between β and γ. The motion out and back takes equal times, so

$$T = 2\sqrt{\frac{ma\sin\alpha}{\lambda}}\int_\beta^\gamma \frac{\mathrm{d}\theta}{\sqrt{(\sin\alpha - \sin\beta)^2 - (\sin\theta - \sin\alpha)^2}}, \qquad \gamma = \arcsin(2\sin\alpha - \sin\beta).$$

(b) $\sin\beta < 2\sin\alpha - 1$**.** Now $2\sin\alpha - \sin\beta > 1 \geq \sin\theta$, so $\dot\theta$ never vanishes for θ between β and $\frac{1}{2}\pi$: the beads pass the ends of a diameter while still moving. After that the chord shortens again. The energy equation involves θ only through $\sin\theta$, which is symmetric about $\frac{1}{2}\pi$, so the motion from $\frac{1}{2}\pi$ to $\pi - \beta$ mirrors the motion from β to $\frac{1}{2}\pi$. The beads stop at $\theta = \pi - \beta$, where the spring is again compressed to its original length. It then pushes them back the way they came. (At $\theta = \pi - \beta$ the equation of motion gives $\ddot\theta < 0$, since $\sin\theta - \sin\alpha < 0$ and $\cos\theta < 0$.) A full oscillation is $\beta \to \pi - \beta \to \beta$, which is four copies of the quarter from β to $\frac{1}{2}\pi$:

$$T = 4\sqrt{\frac{ma\sin\alpha}{\lambda}}\int_\beta^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{(\sin\alpha - \sin\beta)^2 - (\sin\theta - \sin\alpha)^2}}.$$

Since $\sin\alpha < 1$ we have $2\sin\alpha - 1 < \sin\alpha$, so both cases are consistent with $\beta < \alpha$. Case (b) can only happen if $2\sin\alpha - 1 > 0$, that is if $\frac{1}{6}\pi < \alpha < \frac{1}{4}\pi$.

Checks

- In case (a), the substitution $\sin\theta = \sin\alpha - (\sin\alpha - \sin\beta)\cos u$ turns the integral into $\int_0^\pi \mathrm{d}u/\cos\theta$. When $\beta \to \alpha$ this tends to $\pi/\cos\alpha$, which gives back the period in part (i).
- We simulated the two beads moving independently on the wire with $m = 1.3$, $a = 0.7$, $\lambda = 2.1$. The energy equation held to rounding error and the bisector

stayed fixed. For $\alpha = 0.6$, $\beta = 0.59$ the period was 3.766, against 3.7657 from part (i). For $\alpha = 0.7$, $\beta = 0.4$ (case (a)) it was 4.8847, against 4.8846 from the integral. For $\alpha = 0.75$, $\beta = 0.1$ (case (b)) it was 7.0553, against 7.0553, and θ peaked at $\pi - \beta$ as predicted.

Common slips

- Using $\lambda x^2/(2a)$ for the elastic energy. The denominator is twice the *natural length*, $2 \cdot 2a \sin\alpha$.
- In case (b), stopping the integral at $\theta = \frac{1}{2}\pi$ or claiming the beads stop there. They pass the ends of a diameter with nonzero speed, so the period is $4\int_\beta^{\pi/2}$, not $2\int_\beta^{\pi/2}$.

Beyond STEP

The period integrals in part (ii) cannot be done in elementary terms. They are relatives of the *elliptic integrals* that give the exact period of a simple pendulum. Notice that the small-oscillation period in part (i) is independent of amplitude, while the exact period is not. That contrast between linearised and nonlinear oscillators is a central theme of a first course in classical mechanics. Conservation of angular momentum for a system with internal forces, which fixed the bisector here, is treated carefully in Kleppner and Kolenkow, *An Introduction to Mechanics*.

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